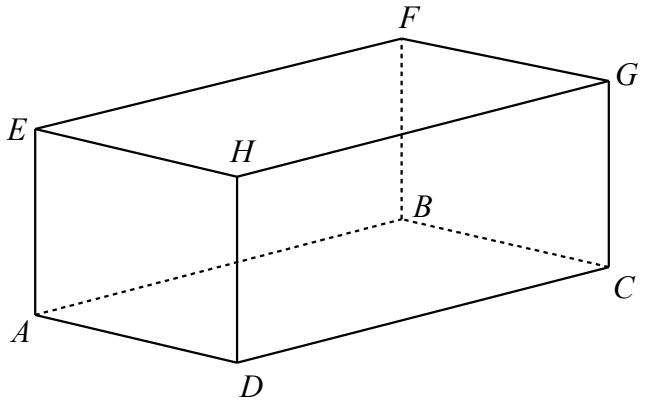
1 The diagram shows a cuboid *ABCDEFGH*.

AE = 4 cm
AD = 5 cm
DC = 8 cm

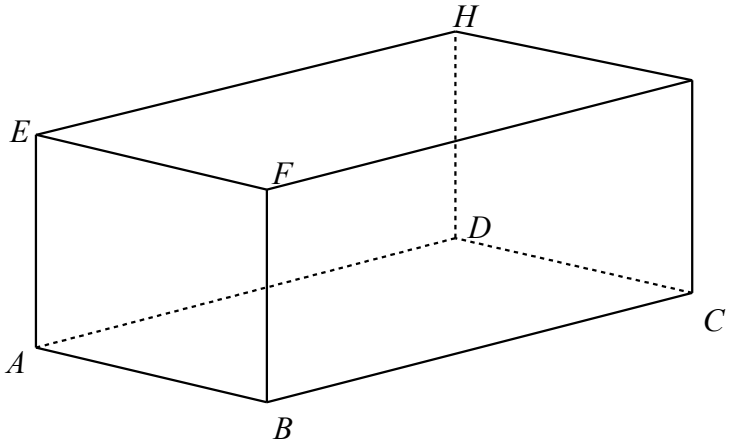


Calculate the length of *AG*.
Give your answer correct to 3 significant figures.

(3 marks)

2 The diagram shows a cuboid *ABCDEFGH*.

AB = 5 cm
AE = 6 cm
AG = 12 cm

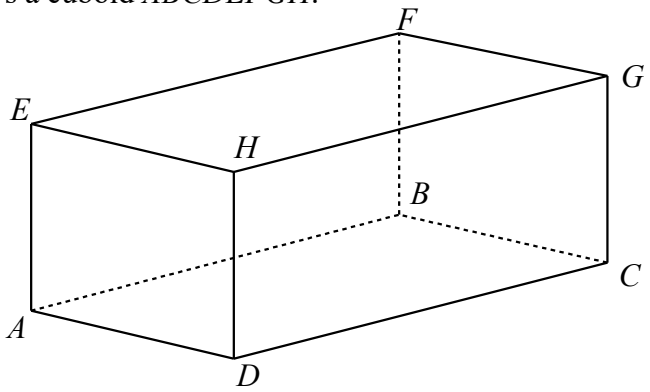


Calculate the length of *AD*.
Give your answer correct to 3 significant figures.

(4 marks)

3 The diagram shows a cuboid *ABCDEFGH*.

AE = 4 cm
AD = 5 cm
DC = 8 cm

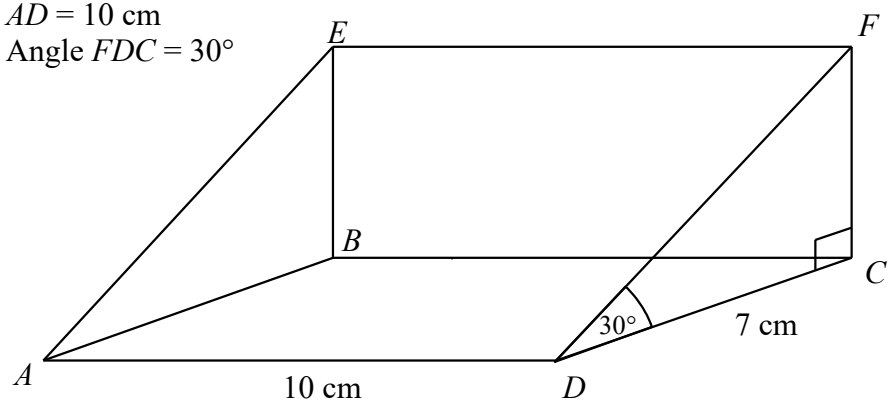


Calculate the size of angle *ECA*.
Give your answer correct to 3 significant figures.

(4 marks)

4 The diagram shows a triangular prism.

CD = 7 cm
AD = 10 cm
Angle *FDC* = 30°



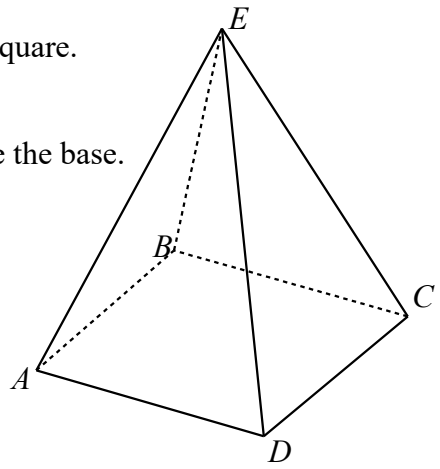
Calculate the size of angle *AFC*.
Give your answer correct to 1 decimal place.

(4 marks)

5 The diagram shows a pyramid.
The base of the pyramid $ABCD$ is a square.

$AB = 5$ cm
The point E is 10 cm vertically above the base.

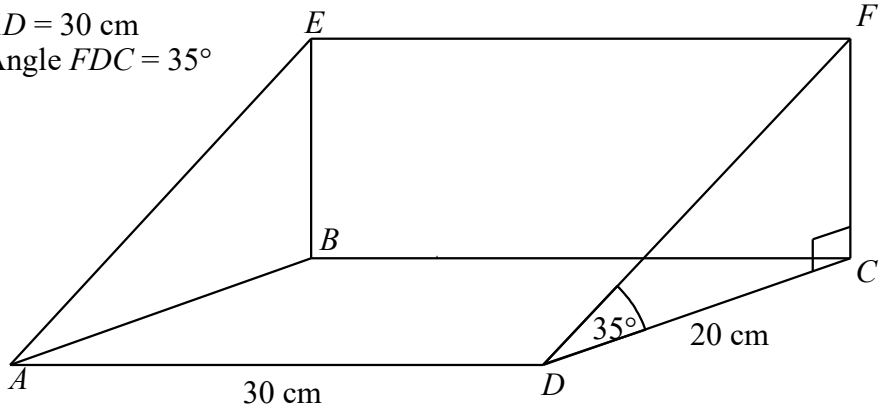
Calculate the size of angle EAC.



(4 marks)

6 The diagram shows a triangular prism.

$CD = 20$ cm
$AD = 30$ cm
Angle $FDC = 35°$

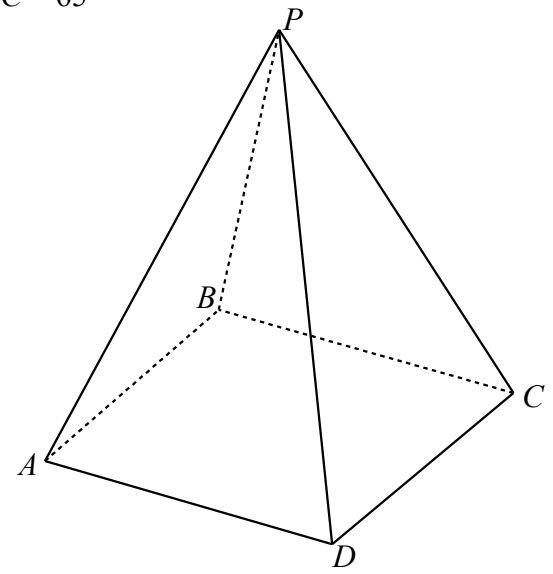


Calculate the size of angle the line AF makes with the plane $ABCD$.
Give your answer correct to 3 significant figures.

(4 marks)

7 The diagram shows a pyramid.
The base of the pyramid $ABCD$ is a square.

$AB = 15$ cm
Angle $PAC = 65°$



Calculate the volume of the pyramid.

(5 marks)